Recognition Technologies Literature Review - Voice Interfaces with Unicomp Visions as an Accessible Aid

Weiser is often credited as curating the original ubiquitous vision, often challenged due to the utopian and naïve nature of his depictions [1]. Current existing Voice User-Interface (VUI) Systems go against this, requiring the user to 'slow-down' and guide technology in order to achieve successful interactions.

1 VUIs within original unicomp visions

Weiser's vision completely removes the notion of a user-interface, and instead enhances locations through the immersion of computers [2]. Weiser dictates a hypothetical day in the life of a corporate woman, Sal. Through this, each monotonous aspect of her routine is enhanced by an invisible device [2]. Despite the idealistic technologies that surround Sal, it could be said that Weiser had comparatively low expectations for VUIs. Perhaps this is since, within Weiser's vision, there is no need for a user to have a conversation with a machine. Instead, the computer should enhance our lives while "[leaving] space for us to connect [..] in essentially human ways." [1] Within his vision the computers are omniscient, predicting Sal's needs to create an interweaved network of devices [2].

Sal wakes in the morning to find her alarm clock brewing her a cup of coffee. This is prompted by her sleep-addled agreement to a simple request. Weiser's visionary device is only able to understand the words "Yes" and "No" [2]. While this may bring doubts to its intelligence, it is likely that Weiser has decided that this is all the device would need to understand. This may have been done to highlight that there is no dire need for overly 'smart' technology that replaces human interaction, though this would be a necessary input method for a screen-less device. A modern reader knows this seamless interaction is unrealistic as a device only capable of understanding minimal responses is unlikely to be beneficial to most users. As sociologist Harvey Sacks comments throughout his lectures, often standardised utterances do not receive the expected responses [3].

2 Current reality of VUIs

The complexities within human-machine conversation may have been avoided by Weiser, but this continues to be a trend within modern technologies. The work of Porcheron et al highlights issues within existing VUIs and uncovers design considerations from a ethnomethodological perspective.

2.1 The need for progressivity and repair

Collaborative work is a common theme: Weiser dreams about devices facilitating the collaboration between colleges by "sharing a virtual office" to create a seamless work-life organisation [2]. While Porcheron's harsh reality focuses on the necessity of users to collaborate to become machine recognisable. Within the presented VUI trials, the user is required to make considerable work to formulate a recognisable utterance that the device is able to understand [4].

Similarly, Fischer forms his own analysis from the transcripts with great consideration to work within Conversational Analysis. The idea of "progressivity" is explored from the observations of Stivers and Robinson who found participants to naturally progress the flow of conversation when halted [5]. Fischer highlights how this desire extends to the those outside the intended recipient. Several of the instances analysed within the paper lead to repeated reformulation of utterances in response to a "non-answer" by the Alexa device. The usual notion of 'turn-taking' is continued, while each member of the party attempts to repair the command and may lead to eventual abandonment by the participants. Prompted from this, Fischer begins to design a framework that prioritises the user's natural desire for "progressivity" as part of his visions for the future of VUIs, as "[halting] interactional progress as the ultimate outcome to avoid" [6].

When presenting his analysis Fischer considers existing design approaches, including that of Reeves who makes the distinction between human conversation and machine interaction [6]. He observes that these are sequenced based as “request and response” pairs. From this, he is able to analyse the design of each “non-answer” response independently, and consider how these provide “resources” for the user to build their utterances upon [7]. In the case of the Alexa device, it is seen that these responses are often not designed in a way that aids the user in repair, relying solely on the desire for progressivity to motivate the interaction. Examples of these can be seen within the transcript in Figure 1 [6].

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01 SUS  Alexa? (0.7) set us a family quiz.
02      (2.5)
03 ALE  sorry. (.) I can't find the answer to the question
        I heard
04      (0.4)
05 EMM  Alexa:? (1.0) Set (0.3) a family quiz
06      (2.3)
07 ALE  sorry. (.) I don't have the answer to that question.
08      (0.4)
[transcript continues in Case 3a]
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Figure 1: Transcript of case 2A. From “Progressivity for voice interface design” by Joel E. Fischer et al.

Current HCI is restricted by the lack of ethnomethodological design: the success of the experience is not only dependent upon the ability to recognise the user’s intentions, but how the device can lead the user through the conversation [7]. Stivers and Robinson make the observation that “accounts are often offered for non-answers” [5], the lack of consideration for this within the Alexa device could be seen as the key breaking point within many of the transcribed instances, as often the user does not correctly understand the error they are troubleshooting. Through this analysis the authors uncover their own vision for the future of interface-less technology to ease the work required to become recognisable. [7] [6]

While Porcheron et al analyse device interactions from a conversational viewpoint, it is clear that the smart device is not viewed as an equal conversational partner. Within a footnote of the paper the authors explain how the speakers collectively view the device as “troublesome input” technology, as if it were a puzzle they must work together to solve [6]. However, there are instances where human nature personifies these devices. For example, within the transcript in Figure 2 [6], the children contribute to the collaborative repair by adding the word “please”. Though it is not clear whether they also feel this instinct of ‘progressivity’ or simply have a desire to join in, this does highlight their natural instinct to repair an utterance by applying previous social experience - overlaying this onto interactions with technology [6].

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01 LIA  Alexa:? (0.9) please set (0.3) a [family quiz.    ]
02 ALL                                   [((laughter))    ]
03      (1.2)
04 ALE  I wasn't able to understand [the question I heard.]
05 EMM                              [ ((laughs))          ]
06 LIA  beep
07      (0.9)
08 CAR  ALEXA, (0.7) FAMily quiz.
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Figure 2: Transcript of case 3A. From “Progressivity for voice interface design” by Joel E. Fischer et al.

3 VUIs as an assistive tool

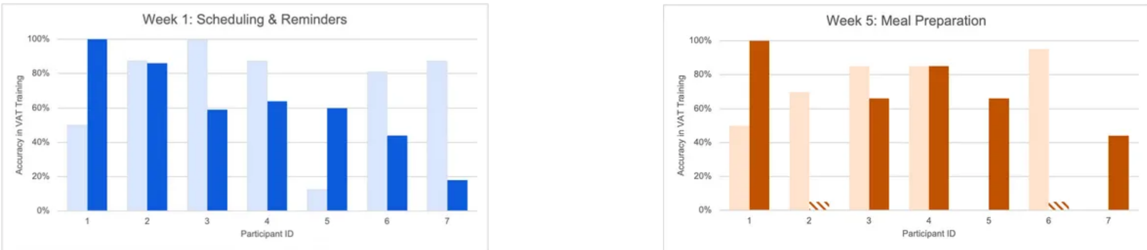
These papers draw attention to how ubiquitous technology should be motivated by the need to decrease work by the user to become recognisable. The removal of a physical interface and traditional input mechanisms should inherently make navigating a device easier, however in reality we must consider who does this aid and who will be left behind.

A review into existing academic research was conducted by Esquivel et al, to explore the use of voice interfaces as an assistive device for those with disabilities. This uncovers the most frequent tasks performed within research studies, and the common barriers that get highlighted. Esquivel observes that while there is a potential use-case for VUI devices, these types of users are often not directly considered when designing the system. She stresses the potential positive impact VUIs could have by aiding in the accessibility of existing technologies, and presenting new tools to help improve their sense of autonomy and independence [8].

Within the ACM Conference on Human Factors in Computing Systems, a research article is presented to evaluate the use of an Alexa device as an assistive technology to aid those with cognitive-communication disorders through necessary daily activities. The authors similarly explain how the recent widespread use of voice-interfaced devices should bring an uptick in accessibility for these users, however this is not always the case due to oversights within the design and a lack of user training [9].

This study was performed amongst a sample of individuals who underwent five weeks of training to observe the external assistance needed in order to achieve successful interactions [9]. While the participants show a clear need for progressivity throughout the trials, collaborative work is implemented by the caregivers, to provide the speaker with verbal guides and visual aids to become more familiar with the device and requirements. The participants are found to require additional prompting and external resources, to achieve the same level of repair reached by those in the previous trials [9].

The author highlights how the participants struggle with the interaction style, requiring assistance with syntactical aspects such as leading with a trigger word, rather than placing it intuitively within a sentence. Many were found to struggle with instinctively verbose and conversational requests rather than succinct commands. This could be due to expected etiquette practices and lack of experience [9]. The findings match the ethnomethodological analysis from the other papers, and how Fischer points out the importance of not viewing the interaction as conversation, in order to avoid these issues [6].



(a) (b)

Figure 3: Accuracy of use with VUIs in first and final week. From "Voice Assistive Technology for Activities of Daily Living: Developing an Alexa Telehealth Training for Adults with Cognitive-Communication Disorders" by Yao Du et al.

Figures 3a,3b [9] show the participants self-reported ability to perform the tasks, and the found "accuracy in executing voice commands". The authors note the discrepancy between the measurements due to participants "acclimating to the device" considering they had no experience prior to the study. There is a general improvement in accuracy as the participants become more familiar with using VUIs. Though at the end of the study not all of the participants were completely independent, the authors describe an overall large increase in confidence and many "[generating] spontaneous commands independently." [9].

These studies show how, although voice-processing is the most common implementation to enable user-centred "interface-less computing"; this has a long way to go to compete with Weiser's visions.

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